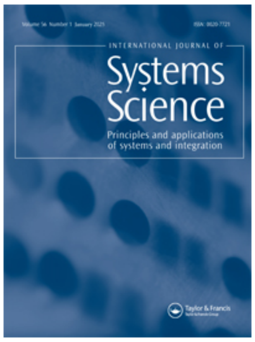

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Fixed/predefined-time stability and designing of terminal sliding mode control of impulsive dynamical systems

Arnab Mapui, Md Arzoo Jamal and Santwana Mukhopadhyay

Department of Mathematical Sciences, Indian Institute of Technology (BHU), Varanasi, India

ABSTRACT

The contribution of the present work is two-fold. Firstly, some results on the predefined and fixed-time stability of non-linear impulsive systems are derived. Impulse-dependent sufficient conditions are derived for predefined-time estimation of settling-time under the influence of stabilising impulses. In contrast, a fixed-time estimate of the settling-time is obtained for destabilising impulses. Secondly, terminal sliding mode (*TSM*) control is designed in the presence of hybrid (continuous and impulsive) matched disturbances for *n*-dimensional systems. It shows that the trajectories can reach the sliding surfaces in fixed-time and thus will stay on it after that under the influence of the proposed *TSM*. Thus, the system states will converge to the origin in a fixed-time. Finally, three examples, which also consist of the stability of Cohen-Grossberg BAM neural networks, are provided to justify the proposed results numerically.

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1. Introduction

Impulsive systems are a class of hybrid systems consisting of both continuous and discrete parts. The continuous part is responsible for the evolution of the states of the system, whereas the discrete portion refers to some sudden changes in the states at specific discrete points. Impulsive systems have drawn serious attention from researchers in recent years because of their applications in several areas, including those in robotics, economics, biology (see Chen et al., 2019; Dashkovskiy & Feketa, 2017; Grizzle et al., 2001 and references therein), and many more. The study of impulsive systems can generally be divided into the impulsive disturbance problem (*IDP*) and impulsive control problem (*ICP*). The case where a given system without impulse possesses specific performances, such as periodic solution, stability, and boundedness, and these can be preserved when the system is subject to sudden disturbances are regarded as *IDP*. Further, if a system without impulses does not possess a particular phenomenon but possesses it via a specific impulsive control, then that is regarded as *ICP*. The main advantage of *ICP* is that the controller acts on the systems at specific instants, drastically reduces the control cost and removes the controller's continuous dependency on the states. From the perspective of impulsive magnitude, *ICP* is the class of stabilising impulses that are beneficial for the systems. In contrast, *IDP* is the class of destabilising impulses that destroy the system's dynamics. Recently, numerous works have been produced concerning *ICP* and *IDP*; some of the articles on *ICP* can be found in Dashkovskiy et al. (2012) and Z. Yang and Xu (2007), while (Jamal et al., 2023; X. Yang et al., 2017) includes some work on *IDP*.

Additionally, much research has also been carried out in the literature (Ambrosino et al., 2009; Arzoo Jamal et al., 2022; Dashkovskiy & Mironchenko, 2013; X. Li et al., 2019, 2020; Naghshtabrizi et al., 2008) on the stability properties of impulsive systems. For example, Naghshtabrizi et al. (2008) studied the exponential stability of time-varying impulsive systems; in X. Li et al. (2020), uniform stability and global asymptotical stability of impulsive systems using event-triggered control are studied. It should be noted that the convergence time of the trajectories to the origin is infinite in these cases, which may not be adequate for many practical applications. To overcome this drawback, Bhat and Bernstein (2000) coined the idea of

CONTACT Arnab Mapui arnabmapui.rs.mat20@itbhu.ac.in Department of Mathematical Sciences, Indian Institute of Technology (BHU), Varanasi 221005, India

finite-time stability (*FTS*). Since the development of the concept of *FTS*, several works on *FTS* of the impulsive systems are reported in X. Li et al. (2019), Ambrosino et al. (2009), Nersesov and Haddad (2008) and Amato et al. (2013); X. Yang and Li (2021). Nersesov and Haddad (2008) provided sufficient conditions for *FTS* of the impulsive dynamical systems along with the design of robust hybrid finite-time stabilising controllers, whereas (Ambrosino et al., 2009) provided *FTS* of linear impulsive systems corresponding to state-dependent impulses. X. Li et al. (2019) discussed some sufficient Lyapunov conditions for *FTS* with a settling-time estimation of impulsive systems for stabilising and destabilising impulses. Despite the advantage of finite time convergence, it suffers from the drawback of the dependency of the settling-time on the initial conditions. So, for different initial conditions one will have different settling-time estimates. Also, in real life, all the initial states used to be unavailable to measure, or they used to be mismeasured. Thus it will lead to poor estimation of the settling-time. To overcome these drawbacks, the concept of fixed-time stability (*FXTS*) was coined in Polyakov (2011). Here, the settling-time used to be independent of initial conditions; rather, it depends only on the system's parameters. Several researches on *FXTS* of impulsive systems can be found in Arzoo Jamal et al. (2022), H. Li et al. (2017) and Hua et al. (2022). For instance, authors in H. Li et al. (2017) studied the *FXTS* of the impulsive dynamical systems corresponding to stabilising impulses by considering the average impulsive interval scheme. In contrast, the authors in Arzoo Jamal et al. (2022) put forward some adequate conditions for *FXTS* of impulsive systems using average impulsive interval and Lyapunov functions for both stabilising and destabilising impulses using linear feedback regulation. Although *FXTS* have several advantages over *FTS*, one can not obtain arbitrarily small convergence time by virtue of tunable parameters. Hence, the concept of predefined-time stability (*PDTS*) was coined in Jiménez-Rodríguez et al. (2017) and elaborated in Jiménez-Rodríguez et al. (2020) and Mapui et al. (2024). Recently, Wei and Li (2023) discussed *PDTS* of impulsive dynamical systems, which gives a non-conservative estimate of the upper bound of the settling-time by considering hybrid impulses and using dwell-time conditions. However, *FXTS* and *PDTS* in the field of impulsive systems are still young and need more attention.

In the literature, several modern control strategies, such as Sliding Mode Control (*SMC*) (Anjum et al., 2022; Y. Liu et al., 2015; Shtessel et al., 2014; Utkin, 1977), Proportional-Integral-Derivative (*PID*) control (Borase et al., 2021), Model Predictive Control (*MPC*) and H-infinity control (Iqbal, 2019), have been extremely effective in studying different models on various fields. However, since the introduction of *SMC* by Utkin (1977), it has been an indispensable control strategy in the field of control engineering due to its high robustness against external bounded disturbances and simplicity of use. It typically consists of two parts: the first involves getting the system to a predefined sliding surface, and the second consists of maintaining the system there using discontinuous control. The state variables then move to the origin along the sliding surface. In the past, several applications of conventional SMC have been shown where state variables reach the origin asymptotically despite finite-time convergence of state variables to the sliding surface. Hence, by adjusting the parameters, one can obtain fast convergence to some small neighbourhood around the origin but can never reach the origin in a finite-time. Inspired by the work of Zak (1988), where the fractional powers of the state variables were taken into account to obtain finite-time convergence, the concept of *TSM* control was developed in Venkataraman and Gulati (1993). Moreover, *TSM* can be divided into Finite-time sliding mode (*FTSM*) control and Fixed-time sliding mode (*FXSM*) control. In *FTSM*, the state variables reach the origin in some finite time, i.e. the settling time depends on the initial state values. In contrast, the idea of *FXSM*–where the system's settling time is independent of the state's initial values–was introduced in Cruz-Zavala et al. (2011). Based on these, many results on *TSM* have been studied in recent years (Feng et al., 2002; Moulay et al., 2022; J. Yang et al., 2013; Zhao et al., 2022). For instance, Feng et al. (2002) presented an *FTSM* non-singular control law and applied the concept to rigid manipulators; J. Yang et al. (2013) deals with the construction of *FTSM* control law in the presence of mismatched disturbances. In addition, the authors in Moulay et al. (2022) constructed an *FXSM* law corresponding to second order system in the presence of mismatched disturbances; construction and analysis of *FXSM* control law for a class of Euler-Lagrange systems in the presence of disturbances and uncertain dynamics is done in Zhao et al. (2022). Further, it should be noted that most of the articles mentioned above on *SMC* suffer from the drawbacks of the chattering problem. It basically means that the controllers have to be switched on and off at an infinite speed, which is practically impossible. Several mitigation strategies have been proposed recently (see Ahmad et al., 2023; Feng et al., 2014; Zarma et al., 2018 and references therein). In brief, Feng et al. (2014) and Ahmad et al. (2023) used the integrated signum function in the controller to reduce the chattering effects. For this, they had to pay the price that the disturbance term should be differentiable and the derivative should be bounded. Whereas, Zarma et al. (2018) used the saturation function, a continuous function, instead of the discontinuous signum function to reduce chattering effects. In the meantime, it is pretty essential to mention that the majority of the articles discussed above on sliding mode control techniques are invalid when impulsive disturbances are present in the system due to the discontinuity of the states. Such impulsive disturbances (a class of discontinuous disturbances) may produce several complex system behaviours, such as instability, oscillations, etc. Let us consider the following example to visualize these facts:

Example 1.1: Consider the following two-dimensional system:

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = 0.1\sin(20t) + u \end{cases}, \tag{1}$$

where, $0.1\sin(20t)$ is the external bounded disturbance and u is the control input. Further, let the system (1) suffers from the impulsive disturbances as follows:

$$\begin{cases} x_1(t) = 1.1x_1\left(t^-\right), & t = s_k \\ x_2(t) = 1.2x_2\left(t^-\right), & t = s_k \end{cases}. \tag{2}$$

It has been shown in the article (Feng et al., 2002), that the system (1) reaches the sliding surface $s(t) = x_1(t) + x_2^{\frac{5}{3}}(t) = 0$ in some finite-time under the control input $u = -(\frac{3}{5}x_2^{\frac{1}{3}}(t) + sgn(s(t)))$. Then, it slides along the corresponding sliding surface to reach the origin $(0,0)^T$ in finite time. Figure 1(a) validates it. Meanwhile, if the system (1) suffers from the impulsive disturbances given in (2) with $s_k = 5k, k \in \mathbb{N}$, the system trajectories and the sliding surface are shown in Figure 1(b). It can be observed from Figure 1(b) that the above-mentioned sliding surface is infeasible and the trajectories are unstable.

As a matter of fact, the main challenges with impulsive disturbances in the system are *i*) that the states may not reach the sliding surface in finite time and *ii*) that even if they do, they may leave the surface while experiencing impulsive disturbances. Several studies have been conducted on the implementation of *SMC* law for systems containing state-dependent impulses, such as Osuna et al. (2016); Oza et al. (2018). The main idea behind the aforementioned papers is that both of these *SMC*s are built in such a way that the states can be returned to the sliding surface prior to the execution of the next impulsive point. In recent years, the concept of designing *SMC*s corresponding to time-dependent impulses (*TDI*) has been explored in X. Li and Zhao (2021), Chen et al. (2021), Hua et al. (2023) and He et al. (2022). Specifically, in X. Li and Zhao (2021), the exponential stability is shown for the linear impulsive system with matched disturbances using average dwell-time impulsive criteria; whereas, (Chen et al., 2021) used switching gain and linear matrix inequality techniques. The authors in Hua et al. (2023) showed exponential synchronisation of Recurrent Neural Networks with time-varying delays. It is to be noted that all the studies provided exponential stability under impulsive disturbances using *SMC*. Different from these, He et al. (2022) offered novel *FTSM* control designs for second-order systems, considering both continuous and impulsive disturbances. However, as per the authors' knowledge, designing *FXSM* control law corresponding to hybrid disturbances is yet to be discussed in the literature.

Further, among several neural networks, such as the Hopfeld network (*HNN*), cellular neural network (*CNN*), and Cohen–Grossberg neural network (*CGNN*), *CGNN* has gained the attention of a lot of researchers in the recent past because of its capability of solving non-linear and uncertain problems practically. Recently, *CGNN* has been widely used in several fields, such as associative memory (Zheng et al., 2010), optimisation problem (Takahashi, 1996), etc. However, all the afore-mentioned neural networks are single-layered. In Kosko (1988), a novel neural network, known as bidirectional associative memory (*BAM*) neural network, was proposed. Afterwards, several works can be found in Arik (2005) and B. Liu and Huang (2006) and references therein. The past few decades have seen a significant increase in interest in

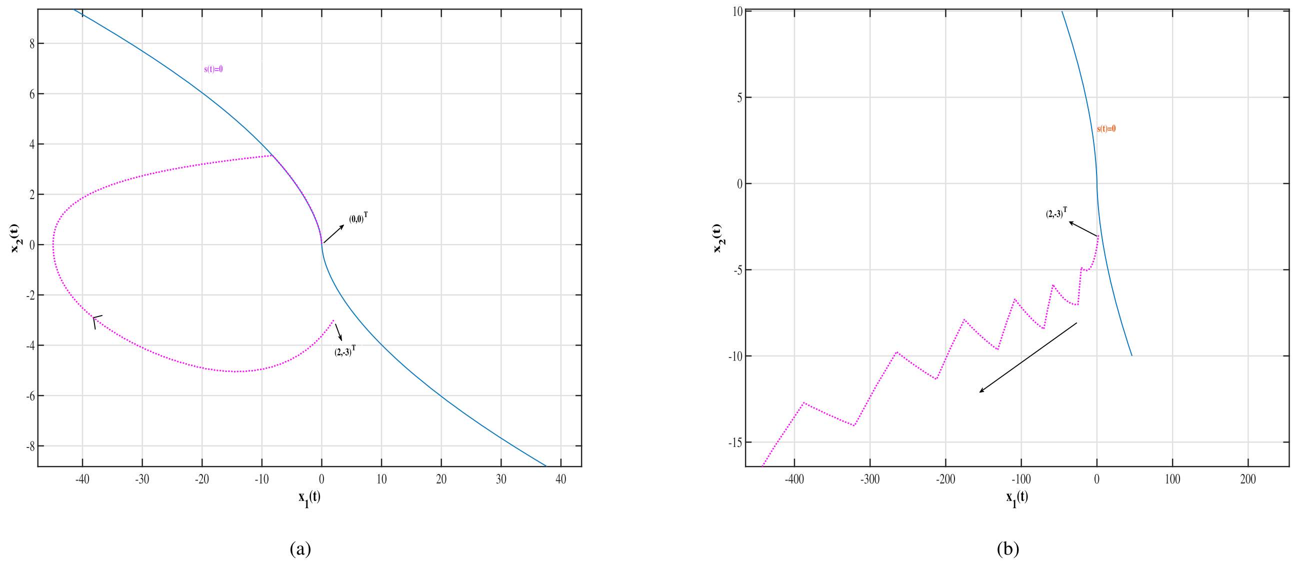


Figure 1. Simulations of Example 1.1 with initial condition $(2,-3)^T$.

CGNNs and *BAM* neural networks due to their wide range of successful applications, including pattern recognition, associative memory, parallel computing, signal and image processing, and more. Meanwhile, some recent results on Cohen–Grossberg *BAM* neural networks with impulses can be found in H. Li et al. (2018) and Jamal et al. (2023). Due to the vast afore-mentioned applications of Cohen–Grossberg *BAM* neural networks, we will apply our theoretical findings to stabilize the Cohen–Grossberg *BAM* neural networks suffering from destabilising impulses in fixed-time with the help of *SMC*.

Motivated by the above discussion, we will study the *FXTS* and *PDTS* of non-linear impulsive systems (*NIS*) using the Lyapunov stability theory in this article. This work further provides a non-singular *FXSM* control law for $n-$dimensional systems with continuous and impulsive disturbances. In summary, the article offers the following contributions:

(1) Sufficient Lyapunov-like conditions for *PDTS* corresponding to stabilising impulses and *FXTS* corresponding to destabilising impulses of *NIS* are provided. Additionally, an impulse-dependent settling-time estimate is established for the case of destabilising impulses.
(2) Unlike existing articles, non-singular *FXSM* is designed for n-dimensional systems corresponding to hybrid disturbances. The reachability of the trajectories to the sliding surfaces in a fixed time is established, and they are also made to stay on the sliding surfaces.
(3) Three examples, including Cohen-Grossberg BAM neural networks as one of them, are provided to verify the theoretical claims numerically.

The rest of the article is organised as follows. Section 2 provides some necessary preliminaries such as notations, definitions and some lemmas that are relevant to the present work. In Section 3, sufficient conditions for *FXTS* and *PDTS* of *NIS* are provided. It also consists of construction of sliding surfaces and the corresponding controllers. Then, in Section 4, numerical verifications corresponding to the theoretical claims are shown. Finally, Section 5 concludes the article with some important highlights.

2. Preliminaries

2.1. Notations

- $\mathbb{R}$ is the set of real numbers; $\mathbb{R}_+ = \{x \in \mathbb{R} : x > 0\}; \mathbb{R}_{\geq 0} = \mathbb{R}_+ \cup \{0\}; \bar{R}_+ = \mathbb{R}_+ \cup \{\infty\}$.
- $\mathbb{Z}^+$ denotes the set of non-negative integers.
- $\mathbb{N}$ is the set of natural numbers.
- $\mathbb{R}^m$, for $m \in \mathbb{N}$, denotes the $m-$dimensional real vector space equipped with Euclidean norm $|\cdot|$. For any $x \in \mathbb{R}^m$, x^T denotes the transpose of x.
- For $x \in \mathbb{R}$, $sign(x)$ denotes the signum function and $sat(x)$ denotes the saturation function, respectively.

- For $\alpha_0, \beta_0 > 0$, the Gamma Functions and Beta Functions are defined as $\Gamma(\alpha_0) = \int_0^{\infty} x^{\alpha_0 - 1} \mathrm{e}^{-x}\,\mathrm{d}x$ and $B(\alpha_0, \beta_0) = \int_0^1 x^{\alpha_0-1}(1-x)^{\beta_0-1}\,\mathrm{d}x$, respectively. Further, for $r \in \mathbb{R}_{\geq 0}$ define Incomplete Gamma Function and the regularised Incomplete Gamma Function as $\gamma(\alpha_0, r) = \int_0^r x^{\alpha_0-1}\mathrm{e}^{-x}\,\mathrm{d}x$ and $P(\alpha_0, r) = \frac{\gamma(\alpha_0, r)}{\Gamma(\alpha_0)}$, respectively. Eventually for $0 \leq r \leq 1$, define Incomplete Beta Function as $b(\alpha_0, \beta_0, r) = \int_0^r x^{\alpha_0-1}(1-x)^{\beta_0-1}\,\mathrm{d}x$ and regularised Incomplete Beta Function as $I(\alpha_0, \beta_0, r) = \frac{b(\alpha_0, \beta_0, r)}{B(\alpha_0, \beta_0)}$.
- $B(\alpha_0, \beta_0) = \frac{\Gamma(\alpha_0)\Gamma(\beta_0)}{\Gamma(\alpha_0+\beta_0)}$, and $\Gamma(\alpha_0 + 1) = \alpha_0\Gamma(\alpha_0)$ for $\alpha_0, \beta_0 > 0$.
- For any real number λ_1, define $\lfloor\cdot\rceil^{\lambda_1} : \mathbb{R} \to \mathbb{R}$ by $\lfloor x\rceil^{\lambda_1} = \mid x \mid^{\lambda_1} sign(x)$ for any $x \in \mathbb{R} \setminus \{0\}$. And for $\lambda_1 > 0$, $\lfloor 0\rceil^{\lambda_1} = 0$.
- For any real numbers a_1 and a_2, define $a_1 \wedge a_2 =$ minimum of a_1 and a_2.

2.2. System's description

Consider the following non-linear impulsive system

$$\begin{cases} \dot{x}(t) = f(x(t)), & t \notin S \\ x(t) = g\left(x\left(t^-\right)\right), & t \in S \\ x(0) = x_0 \end{cases} \tag{3}$$

where $x(t) \in \mathbb{R}^n$ is the state vector and x_0 is the initial state of the system (3). $f : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous non-linear function and $f(0) = 0$. $g : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous function satisfying $g(x) = 0$ iff $x = 0$. $S = \{t_i, i \in \mathbb{Z}^+\}$ is a strictly increasing sequence of impulse time on $(0, \infty)$. We denote the subset $t_i, i = 1, 2, \ldots, K$ of S by S_K for later use, where K denotes the number of impulse points, and impulse time sequences satisfy $t_0 < t_1 < \cdots < t_K < \infty$. As a common assumption, assume $x(t_i) = x(t_i^+)$.

This section contains some mathematical preliminaries needed for the rest of the article such as, a variety of notations, definitions of different types of stability, a special class of function and a lemma.

2.3. On different kinds of stability

All the stability notations defined and referred to hereafter are global, so it will not be indicated.

Definition 2.1 (Stability Notations Wei & Li, 2023): The origin of system (3) is said to be,

- ***Finite-time convergent over the class*** S: if there exist a function $T(x_0, \{t_i\}) : \mathbb{R}^n \times S \to \mathbb{R}_{\geq 0}$ such that the solution, $x(t, x_0)$ corresponding to $x_0 \in \mathbb{R}^n$ and $\{t_i\} \in S$, of system (3) exist and is unique for all $t \in [0, T(x_0, \{t_i\}))$, and $x(t, x_0) = 0$ for all $t \geq T(x_0, \{t_i\})$.
- ***Finite-time stable over the class*** S: if it is *Lyapunov stable* and *Finite-time convergent over the class S*.
- ***Fixed-time stable over the class*** S: if it is *finite-time stable over the class S* and $\exists T_{\max} > 0 : T(x_0, \{t_i\}) \leq T_{\max} < \infty$ for all $x_0 \in \mathbb{R}^n$ and any $\{t_i\} \in S$.
- ***Predefined-time stable over the class*** S: if it is *fixed-time stable over the class S* and for any user defined constant $T_Q > 0$, $T(x_0, \{t_i\}) \leq T_Q$ for all $x_0 \in \mathbb{R}^n$ and any $\{t_i\} \in S$.

2.4. Class of κ^1 functions

Definition 2.2 (κ^1 functions Jiménez-Rodríguez et al., 2020): Consider a continuous function $\xi : \mathbb{R}_{\geq 0} \to [0, 1)$ that is strictly increasing. Further, if $\xi(0) = 0$ and $\xi(r_0) \to 1$ as $r_0 \to \infty$ holds, then the function ξ is called a κ^1 function and it is then denoted by $\xi \in \kappa^1$.

Example 2.1 (Jiménez-Rodríguez et al., 2020): Following is a list of examples of the κ^1 functions:

- $\xi(r_0) = 1 - \exp(-r_0)$;
- $\xi(r_0) = \frac{2}{\pi} arctan(r_0)$;
- $\xi(r_0) = \frac{r_0}{r_0+\alpha_0}, \alpha_0 > 0$;
- $\xi(r_0) = P(\alpha_0, r_0), \alpha_0 > 0$;
- $\xi(r_0) = I(\alpha_0, \beta_0, r_0)$ with $\alpha_0, \beta_0 > 0$.

Lemma 2.1: *Let $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ be a continuous, positive definite and radially unbounded function and $\xi \in \kappa^1$ be differentiable in $\mathbb{R} \setminus \{0\}$. Further, if for any $T_c > 0 \,\exists$ tunable parameters $\Theta \in \mathbb{R}^m$ such that the trajectories of* (3) *without impulses satisfy:*

$$D^+(V(x(t))) \leq -\frac{1}{(1-p)T_c}\frac{(\xi(V(x(t))))^p}{\xi'(V(x(t)))} \tag{4}$$

for $0 \leq p < 1$; then the origin of (3) *without impulses is PDTS in predefined-time T_c. Further, if* (4) *is an equality then $\sup\limits_{x_0 \in \mathbb{R}^n} T(x_0) = T_c$.*

The form of the Lyapunov inequality in Lemma 2.1 is studied in Jiménez-Rodríguez et al. (2020), and they have established the result for continuous systems with tunable parameters. So, to obtain *PDTS* for continuous systems, one must have tunable parameters. Now, if the system experiences some impulses before reaching the settling-time, then it may be possible that the settling-time will change. Thus, the first part of the following section extends the idea of Lemma 2.1 from continuous to impulsive systems.

2.5. An insight to impacts of impulses on predefined-time stability

In this part of the article we will consider an example to give some insights to the impacts of rightly constructed impulsive sequences on predefined-time stability.

Example 2.2: Consider the system given in Jiménez-Rodríguez et al. (2020) as follows:

$$\dot{x} = -\lfloor \lambda_1 \lfloor x \rceil^{\lambda_3} + \lambda_2 \lfloor x \rceil^{\lambda_4} \rceil^{\lambda_5}, \tag{5}$$

where $x \in \mathbb{R}$ represnts the systems' state, $\lambda = [\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5]^T \in \mathbb{R}^5$ is the vector of tunable parameters satisfying $\lambda_1, \lambda_2, \lambda_5 > 0$ and $0 < \lambda_3\lambda_5 < 1 < \lambda_4\lambda_5$. Then it has been shown there that the origin of system (5) is predefined-time stable within predefined-time T_C by choosing the parameters as $\lambda_1 = \lambda_2 = \frac{\Gamma(\frac{1}{4})^4}{4\pi T_C^2}, \lambda_3 = 1, \lambda_4 = 3$ and $\lambda_5 = \frac{1}{2}$.

Now, if the parameters are not tunable, then the origin of system (5) is not predifined-time stable. For example let us suppose that, $\lambda_1 = \lambda_2 = \frac{\Gamma(\frac{1}{4})^4}{4\pi}, \lambda_3 = 1, \lambda_4 = 3$ and $\lambda_5 = \frac{1}{2}$ and are fixed. And we want the origin of system (5) to be stabilised within the predefined-time $T_Q = \frac{1}{2}$. Then it is not possible for every initial value, x_0, of the state x. Because, the settling-time function of system (5) satisfies (see Example 2 of Jiménez-Rodríguez et al., 2020)

$$\sup_{x_0 \in \mathbb{R}} T(x_0) = \frac{\Gamma\left(\frac{1}{4}\right)^2}{\left(\frac{\Gamma(\frac{1}{4})^4}{4\pi}\right)^{\frac{1}{2}} \Gamma\left(\frac{1}{2}\right)(3-1)} = 1.$$

Figure 2 validates this claim.

Now let us add a sequence of impulsive class as $S = \{0.1, 0.2, 0.3\}$ and consider the modified system of (5) as follows:

$$\begin{cases} \dot{x} = -\lfloor \lambda_1 \lfloor x \rceil^{\lambda_3} + \lambda_2 \lfloor x \rceil^{\lambda_4} \rceil^{\lambda_5}, & t \notin S \\ x(t) = \frac{1}{4} x\left(t^-\right), & t \in S \end{cases} \tag{6}$$

Then we can see from Figure 2 that, the state x of system (6) reaches the origin within the predefined-time $T_Q = \frac{1}{2}$.

Remark 2.1: By virtue of Example 2.1, one can obtain that, though the continuous system (system (5)) is not predefined-time stable, its modified version (system (6)), obtained through appropriate injection of impulsive sequences along with impulsive strength, can be stabilised within the predefined-time. So, unlike continuous systems, where tunable parameters are required at least, for the predefined-time stability, such requirements are not needed in the case of impulsive systems. But, in this case predefined-time stability is replaced predefined-time stability over the given class of impulsive sequences as mentioned in Definition 2.1. Hence, the construction of such impulsive sequences which could stabilize the impulsive system (3) within the given predefined time, independent of initial conditions, will be dealt with in the first part of the next section.

3. Main results

In this section, we will be establishing some results corresponding to fixed and predefined-time stability of general *NIS* by considering stabilising and destabilising impulses. Further, *FXSM* controllers are designed by taking hybrid disturbances into consideration.

3.1. Fixed and predefined-time stability

This subsection of the manuscript is dedicated to establishing some results on the predefined and fixed-time stability of *NIS*, considering stabilising and destabilising impulses, respectively.

Theorem 3.1: *Let $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ be a continuous, positive definite, radially unbounded function and $\xi \in \kappa^1$ be differentiable in $\mathbb{R} \setminus \{0\}$ such that, the solution $x(t, x_0)$ of (3) satisfies*

$$\begin{cases} \xi\left(V\left(g\left(x\right)\right)\right) \leq m^{\frac{1}{1-p}} \xi\left(V\left(x\right)\right), & t \in S \\ D^+\left(V\left(x\left(t\right)\right)\right) \leq -\dfrac{1}{\left(1-p\right)T_c} \dfrac{\left(\xi\left(V\left(x\left(t\right)\right)\right)\right)^p}{\xi'\left(V\left(x\left(t\right)\right)\right)}, & t \notin S \end{cases} \tag{7}$$

for $x \in \mathbb{R}^n \setminus \{0\}$, $0 < m < 1, T_c > 0$ and $0 \leq p < 1$; then the origin of (3) is FTS over any class of impulse

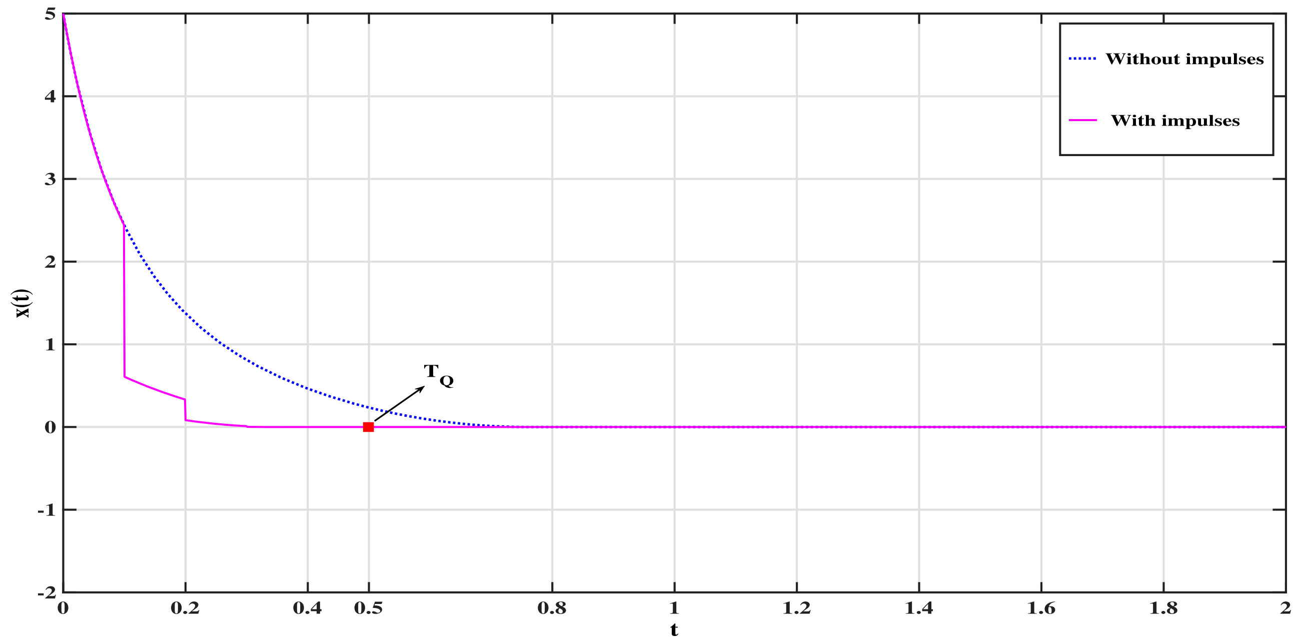


Figure 2. Simulations of Example 2.1 corresponding to the initial value 5.

sequences S. In particular, if for any predefined constant $T_Q > 0$ the impulse sequence $\{t_l\}^K \in S = S_K$ satisfies

$$t_K \le T_c \delta^{K-1} \frac{(\delta - m)}{(1-m)} \xi^{1-p}\left(V\left(x_0\right)\right) < T_c \delta^{K-1} \frac{(\delta - m)}{(1-m)} \tag{8}$$

and

$$K \ge \log_\delta \left(\frac{T_Q}{T_c}\right) \tag{9}$$

where $\delta \in (m, 1)$, then the system (3) is predefined-time stable in predefined-time T_Q corresponding to the class of impulse sequence S_K.

Proof: Without any loss of generality, let $x(t)$ be the solution of the system (3) starting from $(0, x_0)$, where $x_0 \neq 0$, $V_0 = V(x_0)$ and $\xi(V) = \xi(V(x(t)))$. In view of Equation (7), one can easily obtain

$$\xi^{1-p}(V) \le \xi^{1-p}(V_0) - \frac{1}{T_c} t, \quad \forall\, t \in [0, t_1 \wedge T_c).$$

When $t_1 \ge T_c$, i.e. the solution is free from impulses, $\xi^{1-p}(V) \le \xi^{1-p}(V_0)$, $\forall\, t \in [0, T_c]$ and $\xi(V) \equiv 0$, $\forall\, t \ge T_c$. When $t_1 < T_c$, let us consider there are n impulsive points on the interval $[0, T_c]$, i.e. $0 < t_1 < t_2 < \cdots < t_n < T_c$. Since $0 < m < 1$, one can easily obtain that for $i = 0, 1, \dots, n$

$$\xi^{1-p}(V) \le \xi^{1-p}(V_0) - \frac{1}{T_c} t, \quad \forall\, t \in [t_i, t_{i+1}),$$

where $t_0 = 0$ and t_{n+1} is defined by T_c. Hence, $\xi(V) \le \xi(V_0)$, $\forall\, t \in [0, T_c]$ and $\xi(V) \equiv 0$, $\forall\, t \ge T_c$. So, system (3) is *FTS* over any impulsive sequences S. Next, we show that the system (3) will be stabilised in some predefined-time T_Q corresponding to the impulse sequences S_K satisfying Equations (8) and (9). Since $\delta \in (m, 1)$, we have from Equation (8),

$$t_i \le t_K \le T_c \delta^{K-1} \frac{(\delta - m)}{(1-m)} \le T_c \delta^{i-1} \frac{(\delta - m)}{(1-m)}$$

for $i = 1, \dots, K$, i.e.

$$\frac{1}{T_c} \frac{(1-m)}{(\delta - m)} t_i \le \delta^{i-1}. \tag{10}$$

Using Equation (8) one can obtain

$$\xi^{1-p}(V) \le \xi^{1-p}(V_0) - \frac{1}{T_c} t, \quad \forall\, t \in [0, t_1),$$

which, when combines with Equation (10), gives

$$\begin{aligned} \xi^{1-p}(V) &\le \xi^{1-p}\left(V\left(x\left(t_1\right)\right)\right) - \frac{1}{T_c}(t - t_1) \\ &\le m\left(\xi^{1-p}(V_0) - \frac{1}{T_c} t_1\right) - \frac{1}{T_c}(t - t_1) \\ &= m\xi^{1-p}(V_0) + \frac{(1-m)}{T_c} t_1 - \frac{1}{T_c} t \end{aligned}$$

$$\leq m\xi^{1-p}(V_0) + (\delta - m) - \frac{1}{T_c}t$$
$$\leq \delta - \frac{1}{T_c}t \quad \forall\, t \in [t_1, t_2).$$

Proceeding in the similar way, one can acquire that $\forall\, t \in [t_i, t_{i+1})$

$$\xi^{1-p}(V) \leq \delta^i - \frac{1}{T_c}t, \tag{11}$$

where $i = 1, \ldots, K$ and t_{K+1} is defined by $\delta^K T_c$. Now, Equation (11) yields that the settling-time is $\delta^i T_c$. And the claim is that it (the settling-time $\delta^i T_c$) can not exist in the interval $[0, t_K)$. If this is not true, let there be an interval $[t_j, t_{j+1})$, for some $j \in \{1, 2, \ldots, K-1\}$, such that the settling-time $\delta^j T_c \in [t_j, t_{j+1})$.

But from Equation (8) and virtue of $\delta \in (m, 1)$ we have

$$t_{j+1} < t_K < T_c\delta^{K-1}\frac{(\delta - m)}{(1-m)} < T_c\delta^j\frac{(\delta - m)}{(1-m)} < T_c\delta^j$$
$$i.e.\ t_{j+1} < T_c\delta^j,$$

which is a contradiction to the fact that $\delta^j T_c \in [t_j, t_{j+1})$.

Now considering Equation (11) on the interval $[t_K, t_{K+1}]$, one can obtain $\xi^{1-p}(V) \leq \xi^{1-p}(V_0)$ for $t \geq 0$ and $\xi(V) \equiv 0,\ \forall\, t \geq \delta^K T_c$. Hence, the system (3) is *FXTS* with the upper bound of settling-time as $\delta^K T_c$, corresponding to the impulse sequence S_K satisfying Equation (8).

Next, we will be heading towards constructing a sequence of impulses to achieve predefined-time stability corresponding to the system (3). Let $T_Q > 0$ be a predefined constant. And we have to show, $\xi(V) \equiv 0,\ \forall\, t \geq T_Q$.

At this point two cases may arise:

Case 1: $T_Q \geq T_c$. In this case the obtained result $\xi(V) \equiv 0,\ \forall\, t \geq \delta^K T_c$ would take us to our desired result $\xi(V) \equiv 0,\ \forall\, t \geq T_Q$ without considering the extra condition given in Equation (9). Also from the negative definiteness of continuous dynamics of Lyapunov inequality we have, $\xi^{1-p}(V) \leq \xi^{1-p}(V_0)$ for $t \geq 0$. So, the system (3) can be stabilised in predefined-time T_Q in this case.

Case 2: $T_Q < T_c$. In this case the basic idea is to increase the number of impulsive sequences (that is K) such that $\delta^K T_c \leq T_Q$. This is indeed possible beacuse $\delta \in (m, 1)$. Keeping this in mind, from Equation (9) and virtue of $\delta \in (m, 1)$ one can deduce that,

$$T_Q \geq \delta^K T_c. \tag{12}$$

Combining Equation (12) with the fact $\xi(V) \equiv 0,\ \forall\, t \geq \delta^K T_c$, it can be dedeuced that $\xi(V) \equiv 0,\ \forall\, t \geq T_Q$. Also we have, $\xi^{1-p}(V) \leq \xi^{1-p}(V_0)$ for $t \geq 0$. Hence, the system (3) can be stabilised in predefined-time T_Q.

Thus, from both the cases, it can be easily shown that if the impulsive sequence class S_K satisfies Equation (9) along with Equation (8), then the system (3) can be stabilised in predefined-time T_Q. The proof is therefore complete. ■

Remark 3.1: From Theorem 3.1, one can notice that the system (3) without impulses is predefined-time stable in predefined-time T_c, and for that, we need tunable parameters. Now, stabilising impulses are incorporated into the system, and hence, it reduces the settling-time. Let $T_Q > 0$ be any predefined constant, and we are supposed to incorporate the impulses in such a way that the settling-time becomes T_Q. Here, though T_Q is a predefined-time estimate of the settling-time; we do not tune the parameters to obtain that instead, chose the impulsive sequence in a certain way.

Remark 3.2: From the Theorem 1 of article (Hua et al., 2022), one can observe that though the upper bound estimate of settling-time does not include any information of initial states of the system, but to design/estimate the impulse points we still require the information about the initial states of the system. Further, it is mentioned in Remark 4 of Hua et al. (2022) that the impulsive sequence can be estimated without having the information about the initial states but it is only possible when $V(x_0) \leq 1$. However, one can observe in the present work that we do not need such kind of restrictions to obtain fixed-time and predefined-time stability.

Theorem 3.2: *If there exist constants $T_c > 0, m > 1, 0 \leq p < 1$, a positive-definite continuous radially unbounded function $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ and a class κ^1 function ξ such that Equations (7) and (8) hold for any solution $x(t, x_0)$ of system (3), then the system (3) is FXTS over the class S of impulse sequences, where S denotes the sequence class $\{t_l\}$ satisfying*

$$\min\left\{i \in \mathbb{Z}_+ : t_i \geq m^{i-1}T_c\right\} = L_0 < +\infty. \tag{13}$$

Moreover, the settling-time is bounded by

$$T\left(x_0,\{t_l\}\right) \leq m^{L_0-1}T_c. \tag{14}$$

Proof: Let, $x(t,x_0)$ be the solution of system (3) for any initial condition $x_0 \in \mathbb{R}^n \setminus \{0\}$. Then, one can easily obtain from Equation (8) that

$$\xi^{1-p}\left(V\right) \leq \xi^{1-p}\left(V_0\right) - \frac{1}{T_c}t, \quad \forall\, t \in \left[0, t_1 \wedge T_c\right). \tag{15}$$

When $t_1 \geq T_c$, we clearly find from Equation (13) that $L_0 = 1$. Hence, from Equation (15) one obtains $\xi(V) \leq \xi(V_0)$, $\forall\, t \in [0,T_c]$, and $\xi(V) \equiv 0$, $\forall\, t \geq T_c$. Hence, the system (3) is *FTS* in this case. When $t_1 > T_c$, we have $L_0 \geq 2$. From Equation (13) one can obtain that, $t_i < m^{i-1}T_c$ for $i = 1,2,\ldots,L_0-1$ and $t_{L_0} \geq m^{i-1}T_c$. Due to the fact that $m > 1$, we have

$$\begin{aligned}\xi^{1-p}\left(V\right) &\leq \xi^{1-p}\left(V\left(x\left(t_1\right)\right)\right) - \frac{1}{T_c}\left(t-t_1\right)\\ &\leq m\left(\xi^{1-p}\left(V_0\right) - \frac{1}{T_c}t_1\right) - \frac{1}{T_c}\left(t-t_1\right)\\ &\leq m\xi^{1-p}\left(V_0\right) - \frac{1}{T_c}t,\ \forall\, t \in \left[t_1, t_2 \wedge mT_c\right),\end{aligned}$$

Proceeding in the similar way, one can finally obtain

$$\begin{aligned}\xi^{1-p}\left(V\right) &\leq m^{L_0-1}\xi^{1-p}\left(V_0\right) - \frac{1}{T_c}t\\ &\leq m^{L_0-1} - \frac{1}{T_c}t,\\ &\quad \forall\, t \in \left[t_{L_0-1}, t_{L_0} \wedge m^{L_0-1}T_c\right).\end{aligned} \tag{16}$$

But, from Equation (13), we have $t_{L_0} \geq m^{L_0-1}T_c$ and thus it is clear that $\xi(V) \leq m^{\frac{L_0-1}{1-p}}$ $\forall\, t \in [0, m^{L_0-1}T_c]$. Also, the right hand side of Equation (16) is zero for $t = m^{L_0-1}T_c$ and thus $\xi(V) \equiv 0$ $\forall\, t \geq m^{L_0-1}T_c$. Hence, the settling-time $T(x_0,\{t_l\})$ satisfies Equation (14).The proof of Theorem 3.2 is therefore complete. ■

Corollary 3.1: *For any constant $T_L \geq T_c$, the settling-time corresponding to the solutions $x(t,x_0)$ of system* (3) *satisfies*

$$T\left(x_0,\{t_l\}\right) \leq T_L, \quad \forall\ \{t_l\} \in S,$$

where S is the impulse sequence class satisfying Equation (13) *and*

$$L_0 \leq 1 + \log_m\left(\frac{T_L}{T_c}\right). \tag{17}$$

Remark 3.3: Similar to Remark 3.2, Theorem 2 of paper (Hua et al., 2022) needs information about the initial conditions of the states for estimation of impulsive sequences and to get rid of this, one needs to choose $V(x_0) \geq 1$. In contrast, Theorem 3.2 of the present paper does not need any such kind of restrictions to obtain *FTS*. Further, Corollary 3.1 provides some estimates on maximum number of destabilising impulses, corresponding to Equation (14) to maintain the stability of the system (3) in some required time $T_L \geq T_c$. i.e. if we choose number of destabilising impulses greater than $1 + \log_m(\frac{T_L}{T_c})$, then there may exist some class of sequences and some initial conditions such that the system (3) may not be stabilised within the time T_L.

Remark 3.4: In Corollary 1 of article (Hua et al., 2022), the fixed-time stability of impulsive system under destabilising impulse is obtained without using linear feedback regulation. Similar result can also be obtained here by choosing different forms of ξ (see Jiménez-Rodríguez et al., 2020). Further, the settling-time in article (Wei & Li, 2023) does not depend on the impulsive time sequences whereas it depends on it in the current article. Moreover, different from the results of article (X. Li et al., 2019), where the settling-time estimates and construction of impulsive sequence depend on initial states, here both can be done without any dependency on initial states.

3.2. Sliding surfaces and designing of controllers

In this subsection of the article, some results on the design of the sliding surfaces and corresponding controllers are provided for n-dimensional systems with continuous and impulsive disturbances.

Consider the following n-dimensional impulsive system with continuous matched disturbances

$$\begin{cases}\dot{x}_i = f_i\left(x\right) + g_i\left(x\right)u_i + d_i\left(x\right), & t \neq t_k\\ x_i\left(t_k\right) = c_i x_i\left(t_k^-\right), & t = t_k\end{cases} \tag{18}$$

for $i = 1,2,\ldots,n$, where $x_i, u_i \in \mathbb{R}$ are the system states and control inputs respectively, $f_i(x)$ and $g_i(x) \neq 0$ are smooth functions, $d_i(x)$ are continuous matched disturbances satisfying $|d_i(x)| \leq \Theta_i$ for $\Theta_i > 0$ and $c_i > 1$. $\{t_k \mid k \in \mathbb{Z}_+\}$ is a strictly increasing sequence denoted by S. In brief, the first equation of the system (18) is the continuous dynamics, and it suffers from some discontinuous jumps at certain instances $t = t_k$. Here, c_i is greater than 1 and hence, the discontinuous jumps are the destabilising impulses. Further, it should be noted that $f_i(x)$ and $g_i(x)$ are known terms, wheres $d_i(x)$ is unknown, but the bound Θ_i is known. The main goal here is to design non-singular fixed-time terminal sliding mode controller u_i's to stabilize the origin of system (18) in some fixed-time. For this let us design sliding surfaces as follows:

$$M_i = x_i(t) + \frac{1}{(1-p_i)T_{c_i}} \int_0^t \left(|x_i(\tau)|^{p_i}\right)(1+|x_i(\tau)|)^{2-p_i} sign(x_i(\tau))\, d\tau, \quad (19)$$

and the sliding mode control laws u_i's are designed as

$$u_i = -g_i^{-1}(x)\left(f_i(x) + L_i(x) + \frac{1}{n(1-p)T_c} \frac{\xi^p(V(M))}{\xi'(V(M))} sign(M_i) + \Theta_i sign(M_i)\right) \quad (20)$$

where $0 \le p, p_i < 1$, $T_{c_i}, T_c > 0$, $\xi \in \kappa^1$, $\xi' : \mathbb{R}_{\ge 0} \to \bar{R}_+$ and $L_i(x_i) = \frac{1}{(1-p_i)T_{c_i}}(|x_i(t)|^{p_i})(1+|x_i(t)|)^{2-p_i}$ $sign(x_i(t))$.

Theorem 3.3: *Using the controllers given by* (20), *the state trajectories of system* (18) *will reach the sliding surfaces M_i, in some fixed-time given by*

$$\left(c_l^{1-p}\right)^{L_0-1} T_c.$$

corresponding to a subsequence class S_l of S satisfying $\min\{i \in \mathbb{Z}_+ : t_i \ge (c_l^{1-p})^{i-1} T_c\} = L_0 < +\infty$, *where* $c_l = \max\{c_1, c_2, \ldots, c_n\}$.

Proof: Let us consider the Lyapunov function candidate as $V(M) = |M_1| + |M_2| + \cdots + |M_n|$. When $t \ne t_k \in S_l$, combining Equations (18) –(20) one can obtain

$$\begin{aligned} D^+(V(M)) &= \dot{M}_1 sign(M_1) + \dot{M}_2 sign(M_2) + \cdots + \dot{M}_n sign(M_n) \\ &= \sum_{i=1}^n \left\{ \dot{x}_i + \frac{1}{(1-p_i)T_{c_i}} \left(|x_i|^{p_i}\right)(1+|x_i|)^{2-p_i} sign(x_i) \right\} sign(M_i) \\ &= \sum_{i=1}^n \left\{ f_i(x) + g_i(x)u_i + d_i(x) + \frac{1}{(1-p_i)T_{c_i}} \left(|x_i(t)|^{p_i}\right)(1+|x_i(t)|)^{2-p_i} sign(x_i(t)) \right\} sign(M_i) \\ &\le -\frac{1}{(1-p)T_c} \frac{\xi^p(V(M))}{\xi'(V(M))}. \end{aligned} \quad (21)$$

When $t = t_k \in S_l$, one can have

$$\begin{aligned} V(M(t_k)) &= |M_1(t_k)| + |M_2(t_k)| + \cdots + |M_n(t_k)| \\ &= \sum_{i=1}^n \left\{ |x_i(t_k) + \frac{1}{(1-p_i)T_{c_i}} \int_0^{t_k} \left(|x_i(\tau)|^{p_i}\right)(1+|x_i(\tau)|)^{2-p_i} sign(x_i(\tau))\, d\tau| \right\} \\ &\le \sum_{i=1}^n \left\{ |c_i x_i(t_k^-) + \frac{1}{(1-p_i)T_{c_i}} \int_0^{t_k} \left(|x_i(\tau)|^{p_i}\right)(1+|x_i(\tau)|)^{2-p_i} sign(x_i(\tau))\, d\tau| \right\} \\ &\le c_l \sum_{i=1}^n \left\{ |M(t_k^-)| \right\} = c_l V\left(M(t_k^-)\right). \end{aligned} \quad (22)$$

Since, $\xi \in \kappa^1$ is strictly increasing function and $c_l > 1$, we have

$$\xi\left(V\left(M\left(t_k\right)\right)\right) \leq \xi\left(c_l V\left(M\left(t_k^-\right)\right)\right) \leq c_l \xi\left(V\left(M\left(t_k^-\right)\right)\right). \quad (23)$$

Therefore, combining Equations (21) and (23) along with Theorem 3.2, we get that the sliding surfaces M_i will be reached in fixed-time given by $(c_l^{1-p})^{L_0-1}T_c$. This completes the proof. ■

Remark 3.5: From Equation (20), one can note that the proposed controller is discontinuous and suffers from chattering problems. Among several techniques to reduce chattering, one is to replace the discontinuous *sign* function with the continuous saturation function (*sat*). Then, the controller (20) will looks like $u_i = -g_i^{-1}(x)(f_i(x) + L_i(x) + \frac{1}{n(1-p)T_c}\frac{\xi^p(V(M))}{\xi'(V(M))}sign(M_i) + \Theta_i sat(M_i))$. It should be noted that, the control gain will be high in this case.

Now, we need to show that the trajectories will remain on the sliding surfaces at the impulsive instants. Since we have $M_i = 0$ for each i, $\exists k$ such that $\sum_{i=1}^n \mid M_i(t_k^-) \mid = 0$. From Equation (22), one can obtain $\sum_{i=1}^n \mid M_i(t_k) \mid = 0$ and hence $M_i(t_k) = 0$ for each i. So, once the trajectories reach the corresponding sliding surfaces, they will remain on it.

Let us focus on the dynamics of the sliding surface. The dynamics of the sliding surface is provided by

$$M_i(t) = 0, \dot{M}_i(t) = 0.$$

which leads to

$$\dot{x}_i(t) = \frac{1}{\left(1-p_i\right)T_{c_i}} \left(|x_i(t)|^{p_i}\right)\left(1 + |x_i(t)|\right)^{2-p_i} sign\left(x_i(t)\right). \quad (24)$$

Theorem 3.4: *The state trajectories of systems* (24) *will converge to origin in some fixed-time given by* $T_{c_l} = \max\{T_{c_1}, T_{c_2}, \dots, T_{c_n}\}$.

Proof: Considering the Lyapunov function candidate as $V_1 = |x_i|$ and $\xi(V) = \frac{V}{1+V} \in \kappa^1$, we get

$$\begin{aligned}\dot{V}_1 &= \dot{x}_i sign\left(x_i\right) \\ &= -\frac{1}{\left(1-p_i\right)T_{c_i}}\left(|x_i(t)|^{p_i}\right)\left(1 + |x_i(t)|\right)^{2-p_i} \\ &= -\frac{1}{\left(1-p_i\right)T_{c_i}}\frac{\xi^{p_i}(V)}{\xi'(V)}\end{aligned}$$

Hence, by Lemma 2.1, the trajectory of system (24) will converge to the origin in time T_{c_i}. So, the dynamics of the sliding mode for $i = 1, 2, \dots, n$ will converge to zero in the fixed-time and the settling-time is estimated as T_{c_l}.

Combining Theorem 3.3 and Theorem 3.4 we conclude that the system (18) will converge to the origin in fixed-time given by $(c_l^{1-p})^{L_0-1}T_c + T_{c_l}$. ■

Remark 3.6: By virtue of Example 1.1, it should be pointed out that not every sliding surface for continuous systems can be a stand-out performer in the case of impulsive systems. Meanwhile, the authors would like to mention that the construction of the controller (20) is based on the designed sliding surface (19). Also, the controller used to be present in the continuous part of the impulsive systems. Now, the motive of the controller is to make the trajectories reach the sliding surface in spite of the presence of impulsive disturbances in certain instances, which is not trivial as it is in the case of continuous systems. To elaborate this, we would like to get the reader's attention to Figure 1 of the manuscript. From Figure 1(a), it can be seen that the proposed controller is efficient enough to make the trajectories reach the proposed sliding surface when there are no impulsive disturbances. However, the same controller is unable to make the trajectories of the given system to the same sliding surface with the same initial condition when there are impulsive disturbances. This can be verified through Figure 1(b). Further, we would like to mention that it may also be possible that the controller may be able to take the systems' trajectories to the designed sliding surface, but it may leave the sliding upon experiencing destabilising impulsive effects. So, the design of the sliding surface is also vital so that the trajectories do not leave the surface upon experiencing impulsive disturbances. This guarantee is given by the Equation (23).

Remark 3.7: The sliding surfaces and the controllers are designed to make the state trajectories of Equation (18) reach the origin in fixed-time, which is different from any existing article. Also, from Theorem 3.3 and 3.4, one can conclude that the sliding surface and the corresponding controllers are designed

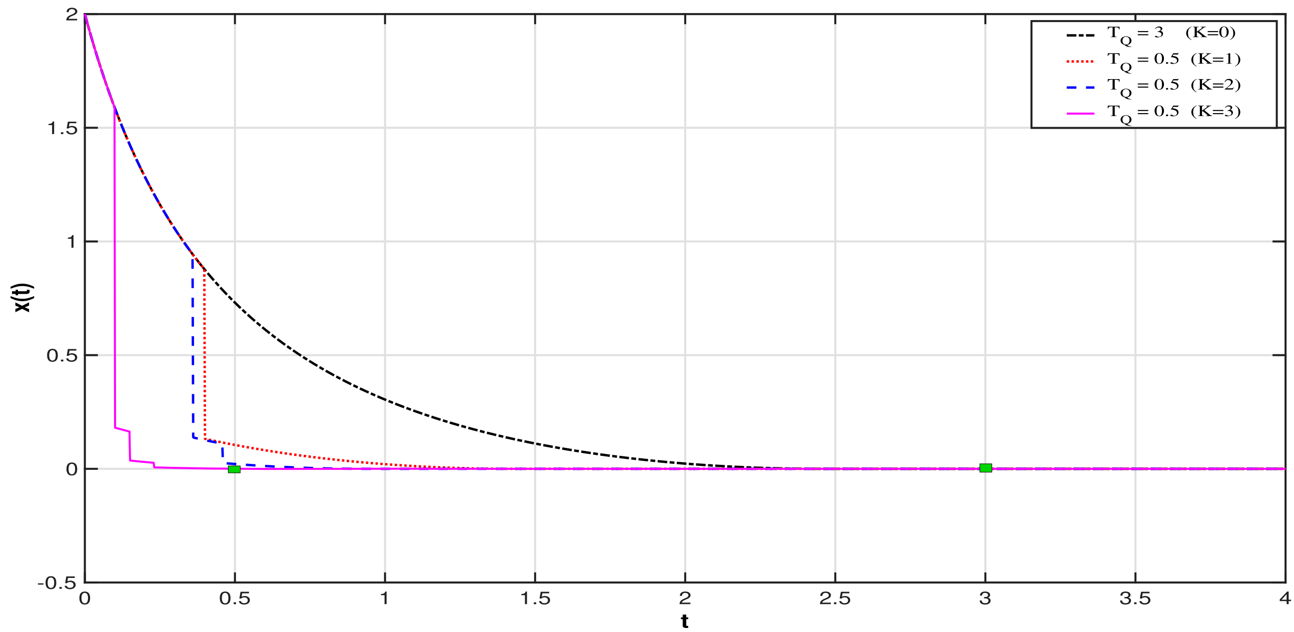


Figure 3. Simulation results of Example 4.1 corresponding to initial value $x_0 = 2$.

for Equation (18), which suffers from both the continuous bounded (may be vanishing or non-vanishing) and destabilising impulses. So, the proposed control strategy is able to withstand the effects of bounded disturbances as well as that of destabilising impulses and make the trajectories of (18) reach the origin within a fixed-time and stay there afterwards.

4. Numerical simulation

Here, we will be providing three examples to verify the aforementioned claims. Further, the simulations are performed using ode45 of MATLAB (R2024a).

Example 4.1: Consider the first-order impulsive system

$$\begin{cases} \dot{x} = -\dfrac{1}{T_c\,(1-p)}\left(|x|^p\right)(1+|x|)^{2-p}\,sign\,(x)\,, & t \neq t_k \\ x\,(t_k) = \dfrac{m|x\,(t_k^-)\,|}{1+(1-m)\,|x\,(t_k^-)\,|}, & t = t_k \end{cases}, \tag{25}$$

where $0 < m < 1, T_c > 0$ and $0 < p < 1$. Then, considering the Lyapunov function candidate as $V(x) = |x|$ and $\xi(V) = \frac{V}{1+V}$ one can deduce that it is of the form of Theorem 3.1 for $p = \frac{1}{2}$. It can be obtained from Lemma 2.1 that the system has settling time T_c when there is no impulsive effects. For simulations, we will be taking $T_c = 3\,s$, $m = \frac{1}{4}$. Figure 3 shows that the state x converges to origin in time $\leq T_c = 3\,s$. Let $T_Q > 0$ be a predefined-time. If $T_Q \geq T_c$ then corresponding to any class of impulsive sequences, the trajectories will converge to origin in time $< T_Q$. Let us now consider the case $T_Q < T_c$. Denote S_K as the class of impulse sequences $\{t_l\}^K$ satisfying Equations (8) and (9), then the system (25) will reach within the time T_Q. For example, considering $\delta = \frac{1}{2}$ and $T_Q = \frac{1}{2}\,s$ one can deduce $t_3 < \frac{1}{4}$ and $K \geq 3$. Figure 3 shows that the state x reaches the origin within the predefined-time T_Q corresponding to the class $S_3 = \{0.1, 0.15, 0.23\}$. Moreover, corresponding to the same conditions if we consider $K = 1$ then $t_1 < 1$. It should be noticed that $K = 1 \ngeq \log_\delta(\frac{T_Q}{T_c})$ and thus from Figure 3 one can observe that the state x is not converging to origin within the predefined-time $T_Q = \frac{1}{2}\,s$ corresponding to the class $S_1 = \{0.4\}$. Similar observation is also done for $K = 2$, in this case $t_2 < \frac{1}{2}$ and the sequence class is choose as $S_2 = \{0.36, 0.46\}$.

Example 4.2: In this example, we will be verifying Corollary 3.1. For this, let us consider a first-order impulsive system as

$$\begin{cases} \dot{x}\,(t) = -\dfrac{1}{(1-p)\,T_c}\left(|x|^p\right)(1+|x|)^{2-p}\,sign\,(x)\,, & t \neq t_l \\ x\,(t_l) = cx\,(t_l^-)\,. \end{cases} \tag{26}$$

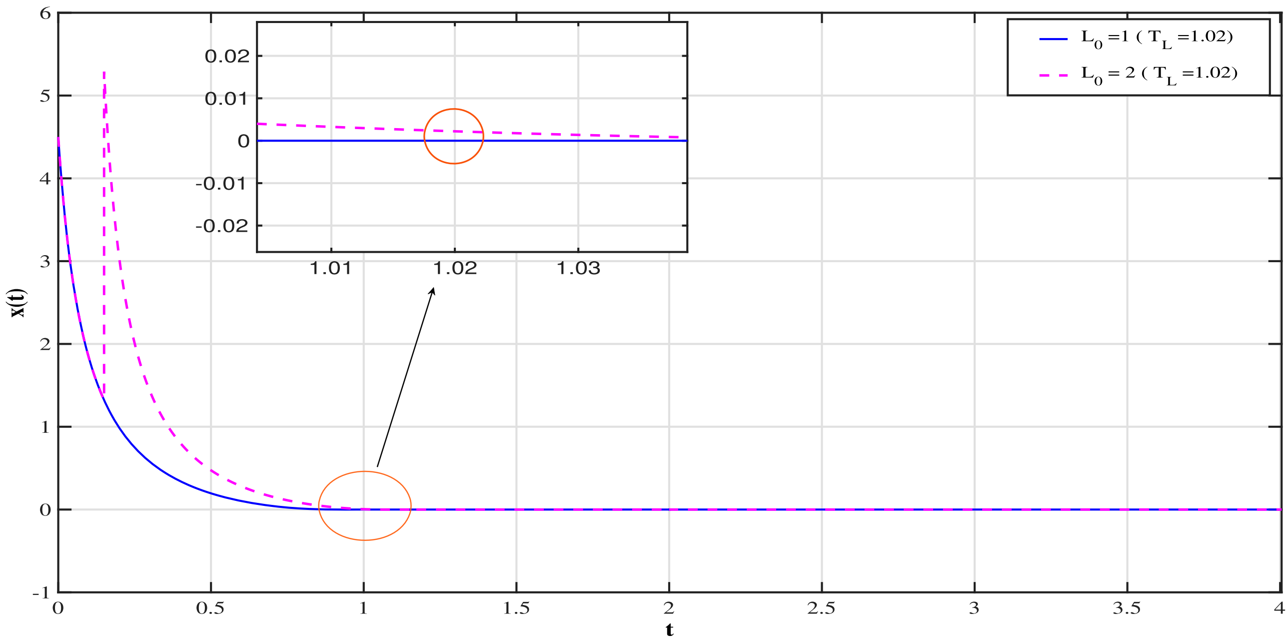
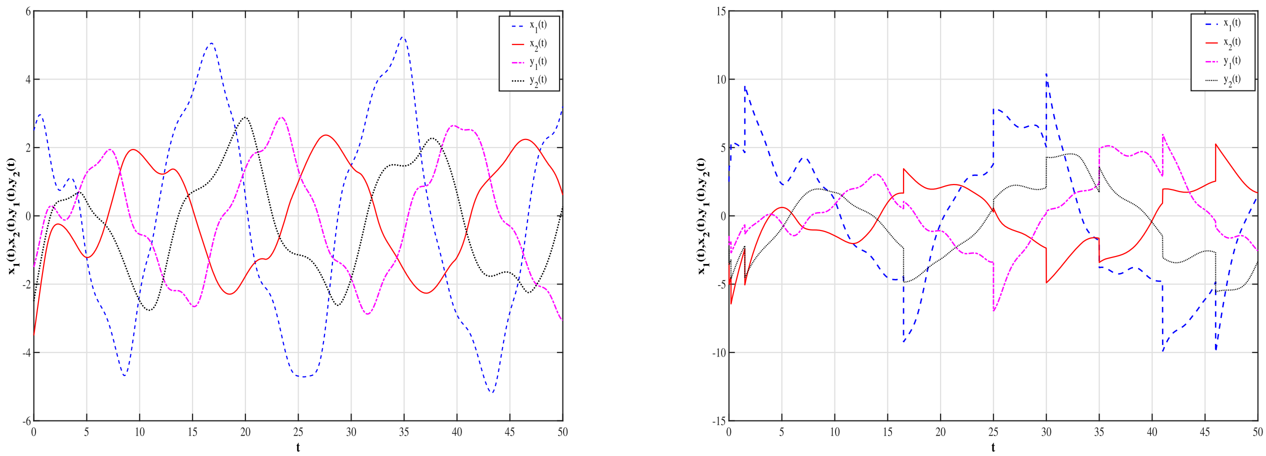


Figure 4. Simulation results of Example 4.2.

And consider the parameters as $p = \frac{1}{2}, T_c = 1, c = 4$. Now, considering the Lyapunov function $V(x) = |x|$ and $\xi(V) = \frac{V}{1+V}$ one obtains that Equation (26) is of the form of Equations (7) and (8) corresponding to $m = 2$. Let S be a sequence of impulses satisfying Equation (13). Considering $T_L = 1.02$ if we have considered $L_0 = 2 \nleq 1 + \log_2(\frac{T_L}{T_C})$ and thus choosing $\{t_l\} = \{0.15, 2.25, 3, 9, \ldots\}$ one can see from Figure 4 that the trajectory is not converging to zero within the time $T_L = 1.02$ corresponding the initial condition $x(0) = 4.5$. However, if we consider $L_0 = 1 \leq 1 + \log_2(\frac{T_L}{T_C})$, then the trajectory will converge to origin within the time $T_L = 1.02$ corresponding to any sequence of impulses and initial conditions, for instance consider $\{t_l\} = \{1.15, 2, 3, 4, \ldots\}$ and initial condition be $x(0) = 4.5$. Figure 4 validates the claims.

Example 4.3: In order to provide a practical application of the aforementioned results (Theorems 3.2–3.4) let us consider the Cohen-Grossberg BAM neural networks with impulse effects. This type of neural network has many applications, such as Pattern recognition, associative memory, parallel computing and many more. Further, every practical system suffers from both continuous and discontinuous disturbances. Under such circumstances, it is thus essential to consider and study the Cohen-Grossberg BAM neural networks experiencing continuous bounded disturbances (may be bounded or unbounded) and discontinuous disturbances (destabilising impulses) as follows:

$$\begin{cases} \dot{x}_u(t) = -a_u(x_u(t))\left\{b_u(x_u(t)) - \sum_{w=1}^{2} k_{uw} f_w(y_w(t))\right\} + v_u + q_u(t), & t \neq t_l, \\ \dot{y}_w(t) = -c_w(y_w(t))\left\{d_w(y_w(t)) - \sum_{u=1}^{2} r_{wu} g_u(x_u(t))\right\} + v_w + p_w(t), & t \neq t_l, \\ x_u(t_l^+) = c_0 x_u(t_l), \quad t = t_l, \\ y_w(t_l^+) = d_0 y_w(t_l), \quad t = t_l, \end{cases} \tag{27}$$



(a) State trajectories of system (27) without control and without impulses.

(b) State trajectories of system (27) without control but with impulses.

Figure 5. State trajectories of system (27) without control. (a) State trajectories of system (27) without control and without impulses. and (b) State trajectories of system (27) without control but with impulses.

where, $u, w = 1, 2, a_1(x_1(t)) = 0.8 + \frac{0.2}{1+x_1^2(t)}, a_2(x_2(t)) = 0.6 + \frac{0.3}{1+x_2^2(t)}, c_1(y_1(t)) = 1.4 + \frac{0.6}{1+y_1^2(t)}, a_1(x_1(t)) = 0.74 + \frac{0.6}{1+y_2^2(t)}$, $b_1(x_1(t)) = 0.5x_1(t)$, $b_2(x_2(t)) = 0.74 x_2(t), d_1(y_1(t)) = 0.2y_1(t), d_2(y_2(t)) = 0.3y_2(t), k_{11} = -2.5$, $k_{12} = 0.5$, $k_{21} = 0.3$, $k_{22} = -1.5, r_{11} = -0.4$, $r_{12} = -1.3$, $r_{21} = 1.5$, $r_{22} = -0.4$, $c_0 = 1.44$, $d_0 = 1.44$, $f_w(y_w(t)) = 0.5(|y_w(t) + 1| - |y_w(t) - 1|)$, $g_u(x_u(t)) = 0.5 \tanh(x_u(t))$, v_u, v_w are the controllers and $q_u(t)$, $p_w(t)$ are the bounded external disturbances bounded by $q_u(t)| \leq \Theta_u$, $|q_w(t)| \leq \Omega_w$. Consider the disturbances as $q_1(t) = 0.5\cos(2t)$, $q_2(t) = 0.5\sin(t)$, $p_1(t) = 0.5\sin(2t)$, $p_2(t) = 0.5\cos(t)$ and thus $\Theta_1 = \Theta_2 = \Omega_1 = \Omega_2 = 0.5$. Now, let us consider the parameters as $p = p_1 = p_2 = p_3 = p_4 = 0.5$ and $T_c = T_{c_1} = T_{c_2} = T_{c_3} = T_{c_4} = 1$. If there were no controllers and no impulses, then the state trajectories of system (27) is shown in Figure 5(a). Figure 5(b) depicts the state trajectories without controllers with destabilising impulsive sequence $\{t_l\} = \{0.01, \ 0.2, \ 1.5, \ \ldots\}$ considering the initial conditions as $[2.5, \ -3.5, \ -1.5, \ -2.5]^T$. From these figures, we can observe that the trajectories are not converging to the origin. Hence, let us construct the controllers to stabilize them to the origin in some fixed-time. For, u, $w = 1, 2$ the sliding surfaces are formed according to Equation (19) as

$$\begin{cases} M_u = x_u(t) + \dfrac{2}{T_{c_u}} \displaystyle\int_0^t \left(|x_u(\tau)|^{0.5}\right)(1 + |x_u(\tau)|)^{1.5} \, sign\,(x_u(\tau)) \, d\tau \\ M_{2+w} = y_w(t) + \dfrac{2}{T_{c_{2+w}}} \displaystyle\int_0^t \left(|y_w(\tau)|^{0.5}\right)\left(1 + |y_w(\tau)|\right)^{1.5} \, sign\,\left(y_w(\tau)\right) \, d\tau \end{cases} \tag{28}$$

and the controllers according to Equation (20). Further, let S be a impulsive sequence satisfying Equation (13), $L_0 = 3$ and thus let us choose $\{t_l\} = \{0.01, 0.2, 1.5, \ldots\}$, $\xi(x) = \frac{x}{1+x}$ for $x > 0$ and let the initial condition be $x(0) = [2.5, -3.5, -1.5, -2.5]^T$. From Figure 6, it can be observed that the trajectories of the system (27) converge to the origin in the fixed time 2.44 *s*. Further, Figure 7 depicts the sliding surfaces and the corresponding states of the system (27).

Remark 4.1: Recently, an article (Jamal et al., 2023) has provided *FXTS* of Cohen–Grossberg BAM neural network under impulsive perturbations using average impulsive interval but they have not considered the continuous disturbances present in the system. Different from that here we have obtained *FXTS* of Cohen–Grossberg BAM neural network under similar parameters by considering hybrid disturbances into account by using sliding mode control.

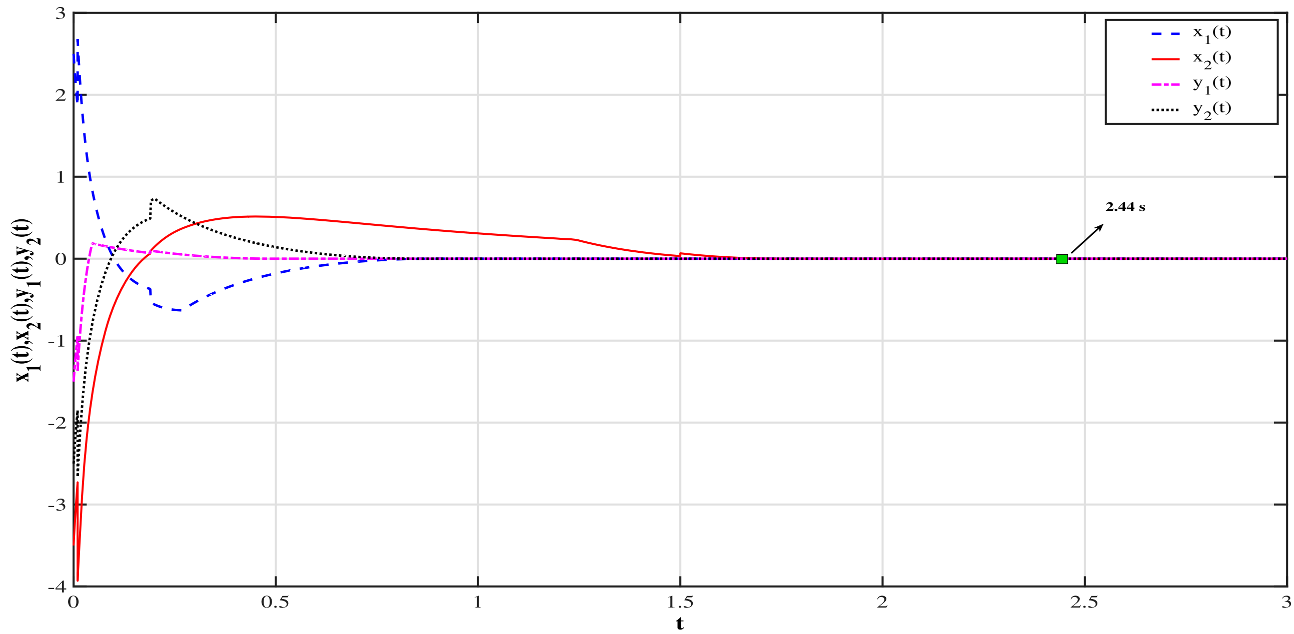


Figure 6. State trajectories of system (27) with control.

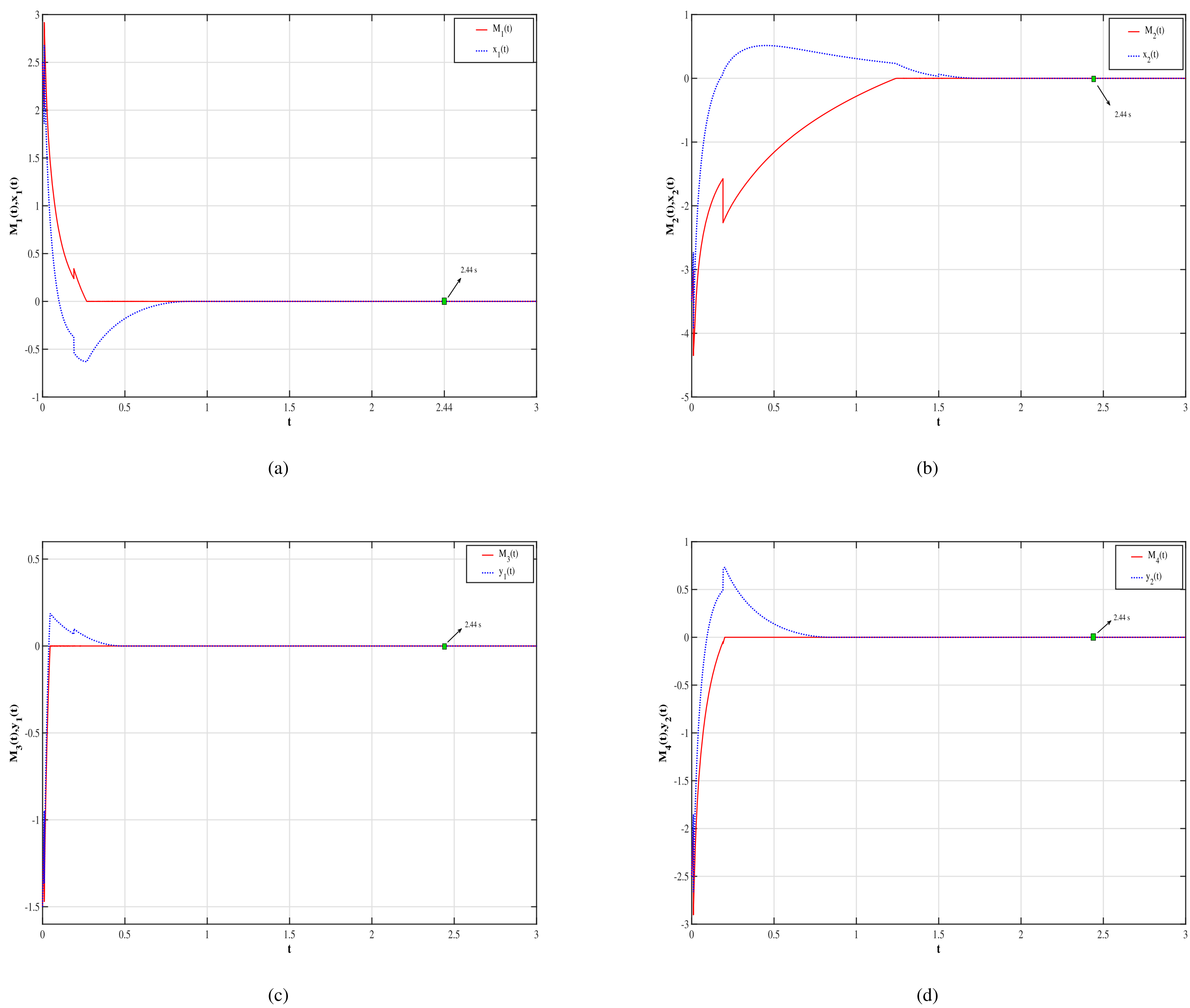


Figure 7. Sliding surfaces and corresponding state trajectories of Example 4.3.

5. Conclusion

In the first part of the article, Lyapunov-like condition for the predefined-time stability of non-linear impulsive systems has been provided by taking only stabilising impulses into consideration. Moreover, it has been noted that predefined-time stability for the case of only destabilising impulses is not possible. Thus, a theorem to ensure fixed-time stability is put forward when the system is affected by destabilising impulses. The inclusion of κ^1 functions helps us to generalize the domain of sufficient Lyapunov-like conditions for predefined and fixed-time stability to a certain extent.

On the other hand, new terminal sliding mode controllers are constructed to stabilise n-dimensional non-linear impulsive systems in the presence of both continuous and impulsive matched disturbances. Different from the results in the existing literature, like (He et al., 2022; X. Li & Zhao, 2021), where the convergence of the trajectories is exponential and finite-time stable, respectively, the convergence of the states to origin is established to be fixed-time in the present work. Further, it can also be seen that the threat of singularity is not present in the control of the current work. Also, numerical validation of the proposed scheme is carried out by considering three examples, one of which is considered as Cohen–Grossberg BAM neural networks.

In the future, further research will be carried out on systems with mismatched disturbances and delayed systems. Further, we will focus on providing some advanced results on chattering reduction for sliding mode control techniques for more practical examples on such systems.

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Disclosure statement

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Data Availability statement

The authors declare that all the data generated and analysed during the study are present in the article.

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